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About

Dr. Chen Jie is currently an Assistant Professor at the Department of Computer Science, Hong Kong Baptist University. Before joining HKBU, he was a senior algorithm engineer at OmniVision Technologies Inc., Singapore. He worked as post-doctoral research fellow at ST Engineering-NTU Corporate Laboratory, Singapore for three years after he received Ph.D. degree in the School of Electrical & Electronic Engineering, Nanyang Technological University, Singapore. He received B.Sc. and M. Eng. degrees both from School of Optical and Electronic Information, Huazhong University of Science and Technology, China. His research focuses on mathematical and machine learning models that solve computational photography and computer vision problems. Dr. Chen has publications in top-tier journals and conferences (TIP, TCI, TCSVT, TGRS, CVPR, ECCV, etc.), and he has been frequently serving as PC/AC members at international academic venues (ACM MM, ICME, VCIP, EUSIPCO etc.). He currently serves as an Associate Editor for the Visual Computer Journal, Springer.

Research Interests

- Computational Photography (Light fields, HDR+, Computational Tomography)
- Computer Vision (Image Synthesis, Scene Understanding, VR/AR applications)
- Novel Imaging Systems and their ISP pipelines

Selected Publications

- Henry Yeung, Junhui Hou, Xiaoming Chen, **Jie Chen**, Zhibo Chen, and Yuk Ying Chung, “Light Field Spatial Super-Resolution Using Deep Spatial-Angular Interleaved CNN,” under major revision in *IEEE Transactions on Image Processing* **(IEEE TIP)**, vol. 28, no. 5, pp. 2319-2330, 2019
- Yun Ni, **Jie Chen**, and Lap-Pui Chau, “Reflection Removal On Single Light Field Capture With Focus Manipulation,” accepted by *IEEE Transactions on Computational Imaging* **(IEEE TCI)**, vol. 4, no. 4, pp. 562-572, 2018.
- **Jie Chen**, Junhui Hou, Yun Ni, and Lap-Pui. Chau, “Accurate Light Field Depth Estimation with Superpixel Regularization over Partially Occluded Regions,” *IEEE Transactions on Image Processing* **(IEEE TIP)**, vol. 27, no. 10, pp. 4889-4900, 2018.
- Henry Yeung, Junhui Hou, **Jie Chen**, Yuk Ying Chung, and Xiaoming Chen, “Fast Light Field Reconstruction With Deep Coarse-To-Fine Modeling of Spatial-Angular Clues,” *European Conference on Computer Vision* **(ECCV)**, 2018.
- **Jie Chen**, Cheen-Hau Tan, Junhui Hou, Lap-Pui Chau, and He Li, “Robust Video Content Alignment and Compensation for Rain Removal in a CNN Framework, ” *IEEE Conference on Computer Vision and Pattern Recognition* **(CVPR)**, 2018.

- **Jie Chen**, Junhui Hou, and Lap-Pui. Chau, “Light Field Denoising via Anisotropic Parallax Analysis in a CNN Framework,” accepted by *IEEE Signal Processing Letters* **(IEEE SPL)**, vol. 25, no. 9, pp. 1403-1407, 2018.
- **Jie Chen**, Junhui Hou, and Lap-Pui Chau, “Light Field Compression with Disparity Guided Sparse Coding based on Structural Key Views,” *IEEE Transactions on Image Processing* **(IEEE TIP)**, vol. 27, no. 1, pp. 314-324, 2018.
- **Jie Chen** and Lap-Pui Chau, “Light Field Compressed Sensing Over a Disparity-Aware Dictionary,” *IEEE Transactions on Circuits and Systems for Video Technology* **(IEEE TCSVT)**, vol. 27, no. 4, pp. 855-865, 2017.
- **Jie Chen** and Lap-Pui Chau, “Multiscale Dictionary Learning via Cross-Scale Cooperative Learning and Atom Clustering for Visual Signal Processing,” *IEEE Transactions on Circuits and Systems for Video Technology* **(IEEE TCSVT)**, vol. 25, no. 9, pp. 1457-1468, 2015.
- **Jie Chen** and Lap-Pui Chau, “A Rain Pixel Recovery Algorithm for Videos With Highly Dynamic Scenes,” *IEEE Transactions on Image Processing* **(IEEE TIP)**, vol. 23, no. 3, pp. 1097-1104, 2014.